

MARKETFLOW ENTERPRISE ERP

System & Prototype Documentation

Application: MarketFlow Enterprise ERP

Version: 2.4.0

Domain: Digital Marketing Operations, Cross-Channel Media Buying & Campaign Governance

Architecture: Modular Client-Side ERP with Dual-Persistence Layer

Database: Supabase PostgreSQL 15 with Row Level Security

Frontend: React 18 + TypeScript + Vite

Live Prototype: <https://marketflow-enterprise.lovable.app>

Repository: <https://github.com/MonishKumarTech/MARKETFLOW-ENTERPRISE>

Contents

SYSTEM INFORMATION & LIVE ACCESS	3
1. PRODUCT OVERVIEW & CORE OBJECTIVE	3
1.1 Product Overview	3
1.2 Core Operational Objectives	3
1. Planning.....	3
2. Pacing	4
3. Attribution	4
4. Action	4
2. SYSTEM ARCHITECTURE & TECHNOLOGY STACK	4
2.1 High-Level Architecture.....	4
2.2 Architecture Components	5
Client Layer	5
Governance & Data Layer.....	5
Cloud Persistence	5
Offline / Local Persistence	5
2.3 Core Technologies.....	5
3. ROLE-BASED ACCESS CONTROL (RBAC).....	6
3.1 Five-Pillar Permission Model.....	6
3.2 Pre-Configured System Personas	6
4. FUNCTIONAL MODULES	6
4.1 EXECUTIVE COMMAND CENTER.....	6
Route: /	6
Key Features.....	7
4.2 PAID CAMPAIGNS & BUDGET PO CONTROL	7
Route: /campaigns	7
Dynamic Custom Fields	7
4.3 CONTENT CALENDAR & CREATIVE WORKFLOW	8
Route: /calendar.....	8

4.4 LEAD ATTRIBUTION & CONVERSION PIPELINE.....	9
Route: /leads	9
4.5 MASTER DATA MANAGEMENT & SCHEMA STUDIO.....	9
Route: /master-admin	9
5. FORMULAS & BUSINESS LOGIC REFERENCE	10
5.1 Cost Per Lead (CPL)	10
5.2 Customer Acquisition Cost (CAC).....	10
5.3 Return on Ad Spend (ROAS).....	10
5.4 Budget Pacing Percentage	11
6. DATABASE ENTITY-RELATIONSHIP OVERVIEW	11
6.1 Core Entities	11
users.....	11
role_permissions.....	11
campaign_master.....	12
campaign_metrics.....	12
content_calendar.....	12
leads.....	12
custom_fields_schema.....	12
custom_field_values	12
6.2 Simplified Database Relationship	13
7. LOCAL SETUP & VERIFICATION.....	13
7.1 Prerequisites	13
7.2 Clone Repository	13
7.3 Configure Environment Variables.....	13
7.4 Start Development Server.....	14
8. SYSTEM VERIFICATION & OPERATIONAL NOTES.....	14
8.1 Authentication Verification.....	14
8.2 RBAC Verification.....	14
8.3 Campaign Verification	14
8.4 Content Workflow Verification.....	15
8.5 Lead Attribution Verification	15
8.6 KPI Verification	15
9. SECURITY & GOVERNANCE CONSIDERATIONS	16
10. END-TO-END BUSINESS FLOW	16
11. CONCLUSION	17
APPENDIX A — QUICK REFERENCE	17
Application Routes	17
Primary KPIs	17
Permission Pillars	18

Technology Summary	18
APPENDIX B — DEMONSTRATION CHECKLIST	18
DOCUMENT CONTROL.....	19
End of Document.....	19

SYSTEM INFORMATION & LIVE ACCESS

Item	Details
Application Name	MarketFlow Enterprise ERP
Version	2.4.0
Live Prototype	https://marketflow-enterprise.lovable.app
Repository	https://github.com/MonishKumarTech/MARKETFLOW-ENTERPRISE
Architecture	Modular Client-Side ERP with Dual-Persistence Layer
Cloud Persistence	Supabase PostgreSQL
Offline Persistence	Local/In-Memory Cache Failover
Target Domain	Digital Marketing Operations
Primary Users	Marketing Managers, Media Buyers, Content Creators, Accountants and Executives

1. PRODUCT OVERVIEW & CORE OBJECTIVE

1.1 Product Overview

MarketFlow Enterprise is a centralized marketing operations Enterprise Resource Planning (ERP) platform designed to eliminate operational data silos across paid advertising, creative asset scheduling, financial purchase-order approvals, campaign monitoring and lead attribution.

The platform provides a unified operational environment where marketing teams can plan campaigns, control budgets, manage content workflows, monitor advertising performance and connect marketing expenditure with business revenue.

1.2 Core Operational Objectives

The system addresses four primary operational requirements:

1. Planning

Schedule and manage multi-platform creative assets across:

- Meta Ads
- LinkedIn
- YouTube
- Other supported marketing channels

2. Pacing

Monitor:

- Approved campaign budgets
- Actual expenditure
- Purchase Order (PO) approvals
- Channel-level spending
- Budget utilization

3. Attribution

Capture and analyze:

- Lead sources
- UTM parameters
- Campaign attribution
- Lead qualification
- Opportunity progression
- Closed customer revenue

4. Action

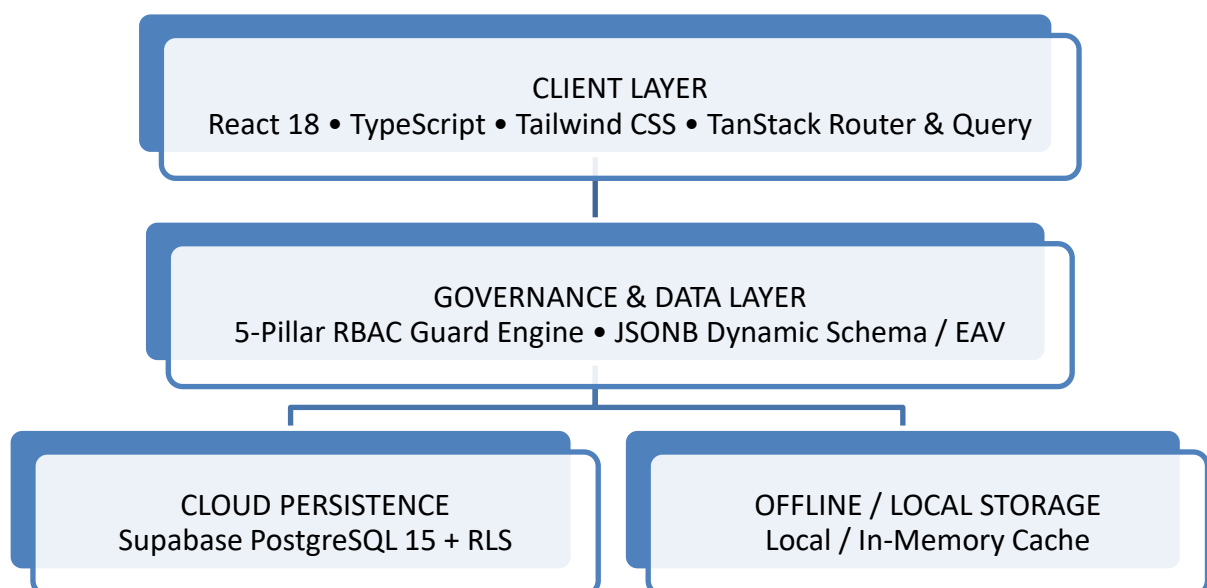
Provide calculated marketing telemetry including:

- Cost Per Lead (CPL)
- Customer Acquisition Cost (CAC)
- Return on Ad Spend (ROAS)
- Budget pacing
- Campaign performance

These metrics support data-driven decisions regarding campaign optimization and budget allocation.

2. SYSTEM ARCHITECTURE & TECHNOLOGY STACK

2.1 High-Level Architecture



2.2 Architecture Components

Client Layer

The client application is responsible for:

- User interface rendering
- Route management
- Form handling
- Data presentation
- Client-side validation
- API/database communication
- Role-aware interface controls

Governance & Data Layer

The governance layer provides:

- Role-Based Access Control
- Permission enforcement
- Dynamic custom fields
- Entity-level access rules
- Database-level security

Cloud Persistence

The primary persistence layer uses:

- Supabase
- PostgreSQL 15
- Row Level Security (RLS)

Offline / Local Persistence

A local fallback layer provides continued operation when cloud persistence is temporarily unavailable or when the application is operating in prototype/offline mode.

2.3 Core Technologies

Technology	Purpose
React 18	Frontend application framework
Vite	Development and build tooling
TypeScript	Static type safety
Tailwind CSS	UI styling framework
Lucide React	Interface icons
TanStack Router	Code-based application routing
TanStack Query	Server-state caching and mutations
Supabase	Backend-as-a-Service platform
PostgreSQL 15	Relational database
Row Level Security	Database-level authorization
JSONB	Dynamic metadata/custom attributes
EAV Pattern	Runtime schema extension

3. ROLE-BASED ACCESS CONTROL (RBAC)

3.1 Five-Pillar Permission Model

Access to routes, records, actions and modal operations is governed through five atomic permissions.

Permission	Purpose
can_view	View records, dashboards, telemetry and analytics
can_create	Create records, assets, campaigns or ad sets
can_edit	Modify existing records and operational parameters
can_delete	Delete records subject to application safeguards
can_approve	Approve creative assets, purchase orders or governed actions

The permission model allows administrators to control access independently for different roles and resources.

3.2 Pre-Configured System Personas

Role	Default Email Identity	Password	Core Capabilities
Workspace Admin	alex.rivera@marketflow.io	marketflow2026	Schema administration, custom fields and RBAC management
Campaign Manager	sarah.j@marketflow.io	marketflow2026	Campaign orchestration, content review, approvals and scheduling
Corporate Accountant	elena.r@marketflow.io	marketflow2026	PO approval, budget control and financial reconciliation
Content Creator	maya.lin@marketflow.io	marketflow2026	Creative drafting and approval submission
Media Buyer	marcus.v@marketflow.io	marketflow2026	Ad monitoring, channel spend pacing and CPL tracking
Executive CMO	aris.thorne@marketflow.io	marketflow2026	Read-only executive analytics and telemetry

Security Note: These credentials are suitable only for a prototype/demo environment. They should not be used as production credentials. Production deployment should use individually provisioned accounts, secure password policies and environment-managed secrets.

4. FUNCTIONAL MODULES

4.1 EXECUTIVE COMMAND CENTER

Route: /

The Executive Command Center provides a consolidated view of marketing performance.

Key Features

KPI Telemetry

Live aggregate metrics include:

- Total Spend
- Attributed Closed Revenue
- Blended CPL
- Blended CAC
- Blended ROAS

Zero-Division Safeguard

Business calculations include conditional safeguards to prevent:

- Division-by-zero errors
- NaN values
- Infinite values
- Uninitialized metric crashes

Cross-Channel Performance

Performance can be analyzed across:

- Meta Ads
 - Google Ads
 - LinkedIn Marketing
 - YouTube
-

4.2 PAID CAMPAIGNS & BUDGET PO CONTROL

Route: /campaigns

This module provides campaign management, budget monitoring and financial governance.

Live Budget Pacing

The system calculates:

$$\text{Budget Pacing} = \left(\frac{\text{Actual Spend to Date}}{\text{Approved Budget Cap}} \right) \times 100$$

Financial PO Gating

Budget changes and governed campaign activations require financial approval before becoming operational.

This provides separation between:

- Campaign planning
- Financial authorization
- Campaign activation

Dynamic Custom Fields

Campaign records can support additional runtime attributes such as:

- Marketing Area Allocation
- Regional Distribution
- Business Unit
- Internal Campaign Classification

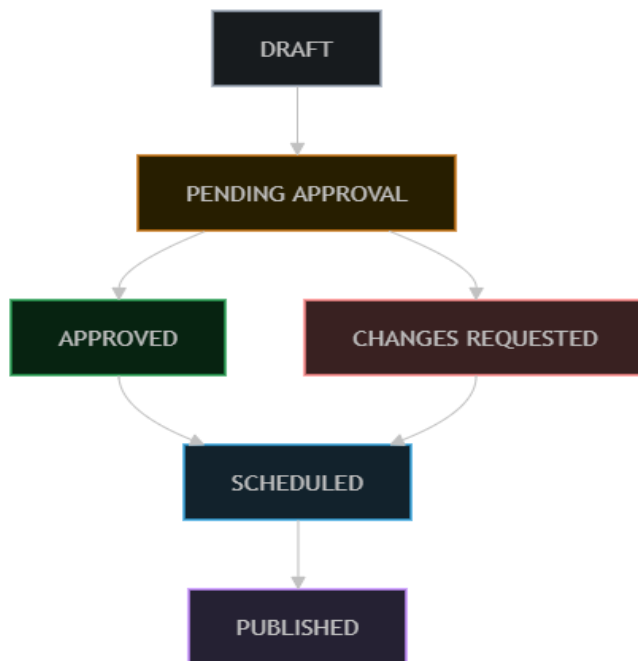
These fields can be introduced through the dynamic schema system without requiring conventional database migrations.

4.3 CONTENT CALENDAR & CREATIVE WORKFLOW

Route: /calendar

The Content Calendar manages creative planning, review and publishing governance.

Content Lifecycle



Role Separation

Content Creator

Can:

- Create drafts
- Edit creative content
- Submit assets for approval

Cannot:

- Approve assets
- Schedule approved assets
- Override approval controls

Campaign Manager / Administrator

Can:

- Review submitted content
- Approve content
- Request changes
- Schedule approved content

This separation provides a controlled content governance process.

4.4 LEAD ATTRIBUTION & CONVERSION PIPELINE

Route: /leads

The Lead Attribution module connects marketing campaigns with lead progression and revenue.

UTM Attribution

Lead records can capture:

- Source
- Medium
- Campaign
- Campaign ID
- Other attribution metadata

Four-Stage Funnel



Closed-Loop Attribution

When a lead reaches:

Customer Won

the associated deal revenue is attributed back to the originating campaign.

This enables campaign-level calculation of:

- Revenue
- ROAS
- CAC
- Conversion performance

The resulting feedback loop connects advertising expenditure to actual business outcomes.

4.5 MASTER DATA MANAGEMENT & SCHEMA STUDIO

Route: /master-admin

The Master Administration module provides centralized configuration and governance.

Staff Workforce Directory

The system provides:

- Personnel records
- Profile management
- Staff information
- Profile photo handling
- Client-side image validation

Profile image validation supports the configured size range:

50 KB – 100 KB

Dynamic Schema Studio

Administrators can define custom attributes including:

- Text
- Number
- Select
- Date

The attributes are linked to supported entities using JSONB-based dynamic storage.

Live RBAC Permission Grid

Administrators can modify permissions using a matrix interface.

Example:

Role	View	Create	Edit	Delete	Approve
Administrator	✓	✓	✓	✓	✓
Campaign Manager	✓	✓	✓	Limited	✓
Accountant	✓	✓	✓	Limited	✓
Content Creator	✓	✓	✓	Limited	✗
Media Buyer	✓	✓	✓	Limited	✗
Executive CMO	✓	✗	✗	✗	✗

The actual permissions remain configurable through the system's RBAC engine.

5. FORMULAS & BUSINESS LOGIC REFERENCE

5.1 Cost Per Lead (CPL)

$$CPL = \frac{\text{Total Paid Advertising Spend}}{\text{Total Ingested Leads}}$$

Purpose

CPL measures the advertising cost associated with generating a lead.

5.2 Customer Acquisition Cost (CAC)

$$CAC = \frac{\text{Total Marketing \& Advertising Spend}}{\text{Total Closed – Won Customers}}$$

Purpose

CAC measures the total marketing investment required to acquire a paying customer.

5.3 Return on Ad Spend (ROAS)

$$ROAS = \frac{\text{Attributed Closed Deal Revenue}}{\text{Total Media Advertising Spend}}$$

Example

If advertising spend is ₹1,00,000 and attributed revenue is ₹4,20,000:

$$\text{ROAS} = (\text{₹}4,20,000) / (\text{₹}1,00,000) = \text{₹}4.2$$

This means ₹4.20 of attributed revenue was generated for every ₹1 spent on media.

5.4 Budget Pacing Percentage

$$\text{Budget Pacing \%} = \left(\frac{\text{Actual Spend to Date}}{\text{Allocated Budget Cap}} \right) \times 100$$

Purpose

Budget pacing measures spending velocity against the approved budget limit.

It helps identify:

- Under-spending
 - Normal pacing
 - Rapid expenditure
 - Potential budget overruns
-

6. DATABASE ENTITY-RELATIONSHIP OVERVIEW

- MarketFlow uses PostgreSQL as its primary relational data store.
- The analytics layer uses pre-aggregation techniques, including Common Table Expressions (CTEs), to calculate spend and revenue independently before joining datasets.
- This approach helps prevent **Cartesian row multiplication** and incorrect aggregate calculations.

6.1 Core Entities

users

Stores:

- System identities
- User information
- Assigned roles
- Authentication-related references

role_permissions

Stores:

- Role
- Resource
- can_view
- can_create
- can_edit
- can_delete
- can_approve

campaign_master

Stores:

- Campaign metadata
- Campaign status
- Budget caps
- Campaign configuration
- Attribution identifiers

campaign_metrics

Stores campaign performance data including:

- Date/time
- Spend
- Impressions
- Clicks
- Channel information
- Other campaign metrics

content_calendar

Stores:

- Creative content
- Channel
- Scheduled date
- Copy
- Approval status
- Publishing status

leads

Stores:

- Prospect information
- UTM source
- UTM medium
- Campaign reference
- Lead stage
- Deal value
- Conversion status

custom_fields_schema

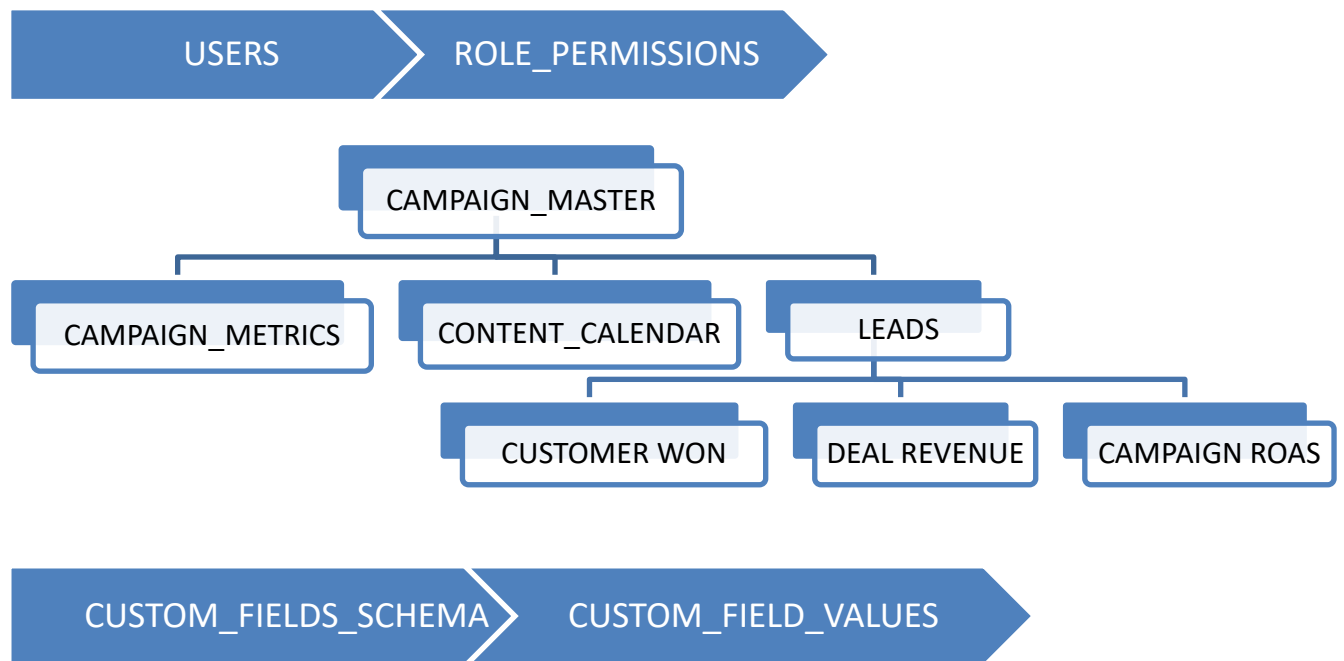
Defines runtime custom fields including:

- Field name
- Field type
- Associated entity
- Configuration
- Validation information

custom_field_values

Stores dynamic entity-specific values using JSONB.

6.2 Simplified Database Relationship



7. LOCAL SETUP & VERIFICATION

7.1 Prerequisites

The development environment should have:

- Node.js
- npm
- Git
- Modern web browser
- Supabase project, if cloud persistence is enabled

7.2 Clone Repository

Open a terminal and execute:

```
git clone https://github.com/MonishKumarTech/MARKETFLOW-ENTERPRISE.git
cd MARKETFLOW-ENTERPRISE
npm install
```

7.3 Configure Environment Variables

Create a .env file in the project root.

VITE_SUPABASE_URL=https://your-project-id.supabase.co

VITE_SUPABASE_ANON_KEY=your-publishable-anon-key

These variables configure the frontend connection to Supabase.

Security: Never commit .env files containing private credentials or secret keys to GitHub. Only publish the appropriate client-side/publishable key intended for frontend use.

7.4 Start Development Server

Execute:

```
npm run dev
```

The Vite development server will normally be available at:

<http://localhost:5173>

Open the address in a modern web browser.

8. SYSTEM VERIFICATION & OPERATIONAL NOTES

8.1 Authentication Verification

Verify that each configured prototype persona can access the system according to its assigned role.

Test:

- Login
 - Logout
 - Role switching
 - Session persistence
 - Unauthorized route access
-

8.2 RBAC Verification

For every major module, verify the five permission pillars:

VIEW
CREATE
EDIT
DELETE
APPROVE

A user should not be able to perform an operation when the corresponding permission is disabled.

8.3 Campaign Verification

Verify:

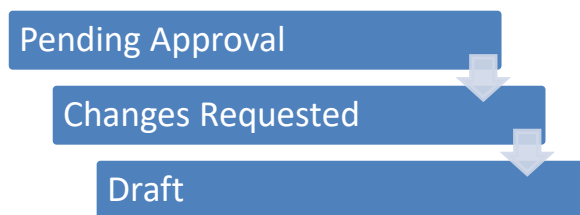
- Campaign creation
 - Budget allocation
 - Spend entry
 - Budget pacing
 - PO approval
 - Campaign activation
 - Dynamic custom fields
-

8.4 Content Workflow Verification

Test the complete workflow:



Also verify the alternate workflow:



8.5 Lead Attribution Verification

Create test leads with different:

- Sources
- Mediums
- Campaigns
- Lead stages
- Deal values

Then verify that a Customer Won lead contributes revenue to its associated campaign.

8.6 KPI Verification

Verify that the Executive Command Center correctly calculates:

- Total Spend
- Closed Revenue
- CPL
- CAC
- ROAS
- Budget Pacing

Also test zero-data conditions to ensure the interface does not display:

NaN
Infinity
Undefined

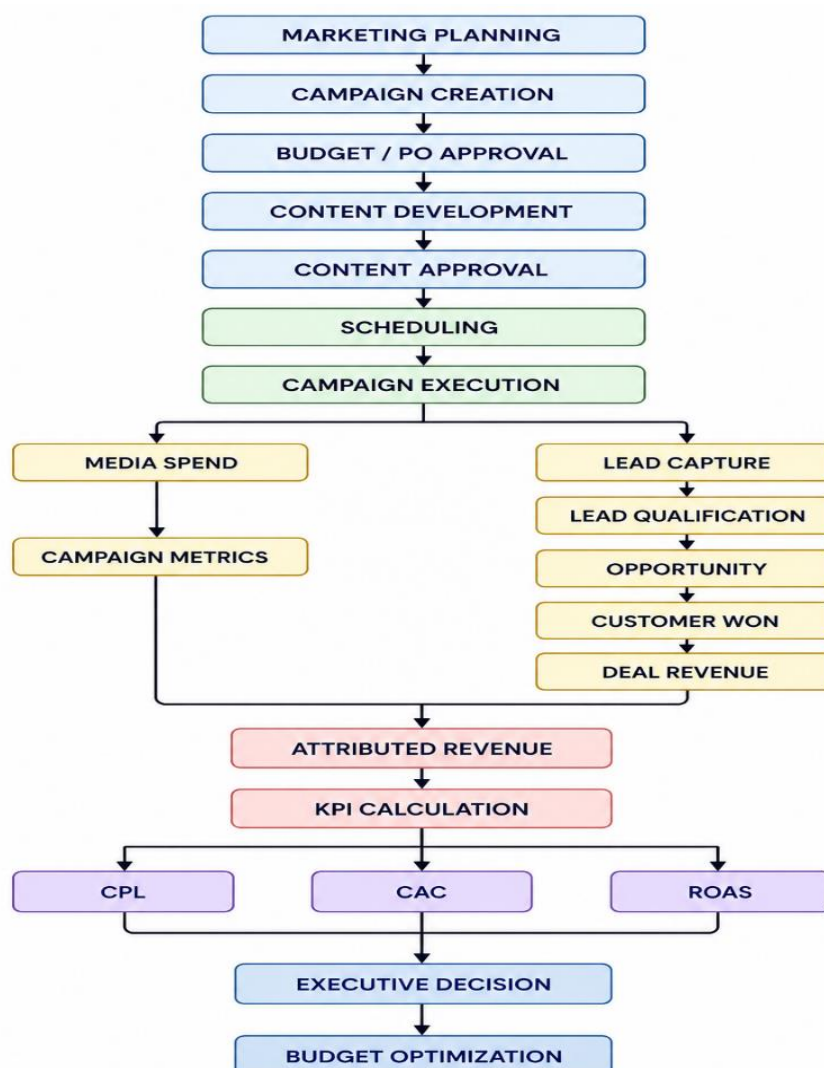
9. SECURITY & GOVERNANCE CONSIDERATIONS

For production deployment, the following controls should be implemented or verified:

1. Replace all demo credentials with individually provisioned accounts.
 2. Enforce strong password policies.
 3. Enable appropriate Supabase Row Level Security policies.
 4. Never expose service-role credentials in frontend code.
 5. Store environment secrets outside source control.
 6. Implement audit logging for sensitive actions.
 7. Restrict administrative operations to authorized users.
 8. Validate all user-supplied data server-side.
 9. Implement backup and recovery procedures.
 10. Review database policies before production deployment.
 11. Apply least-privilege access to every role.
 12. Separate development, testing and production environments.
-

10. END-TO-END BUSINESS FLOW

The overall MarketFlow operating model can be summarized as:



11. CONCLUSION

MarketFlow Enterprise ERP provides a unified operational framework for digital marketing management by connecting campaign planning, creative governance, financial authorization, advertising expenditure, lead attribution and revenue performance.

Its modular architecture combines:

- React-based client architecture
- TypeScript type safety
- Supabase PostgreSQL persistence
- Row Level Security
- Dynamic JSONB attributes
- Five-pillar RBAC
- Campaign governance
- Content approval workflows
- Lead attribution
- Marketing performance analytics

The central design principle is to transform marketing operations from disconnected activities into a **closed-loop, measurable and governed system**.

PLAN → APPROVE → EXECUTE → MEASURE → ATTRIBUTE → OPTIMIZE

This workflow allows MarketFlow to function not merely as a campaign management interface, but as an integrated marketing operations ERP prototype capable of connecting **budget → activity → leads → customers → revenue → decision-making**.

APPENDIX A — QUICK REFERENCE

Application Routes

Route	Module
/	Executive Command Center
/campaigns	Paid Campaigns & Budget PO Control
/calendar	Content Calendar & Creative Workflow
/leads	Lead Attribution & Conversion Pipeline
/master-admin	Master Data & Schema Studio

Primary KPIs

KPI	Formula
CPL	$\text{Paid Ad Spend} \div \text{Ingested Leads}$
CAC	$\text{Marketing \& Ad Spend} \div \text{Closed-Won Customers}$
ROAS	$\text{Attributed Closed Revenue} \div \text{Media Spend}$
Budget Pacing	$\text{Actual Spend} \div \text{Allocated Budget} \times 100$

Permission Pillars

can_view
can_create
can_edit
can_delete
can_approve

Technology Summary

React 18
TypeScript
Vite
Tailwind CSS
TanStack Router
TanStack Query
Supabase
PostgreSQL 15
Row Level Security
JSONB / EAV

APPENDIX B — DEMONSTRATION CHECKLIST

- Application loads successfully
 - Authentication works
 - Role selection works
 - RBAC restrictions work
 - Executive dashboard displays KPIs
 - Campaign creation works
 - Budget pacing calculates correctly
 - PO approval workflow works
 - Content calendar loads
 - Content approval workflow works
 - Scheduling restrictions work
 - Lead creation works
 - UTM attribution works
 - Customer Won updates campaign revenue
 - ROAS recalculates correctly
 - Dynamic custom fields work
 - Staff management works
 - Database persistence works
 - Offline/local fallback works
 - Zero-data KPI safeguards work
 - Production security review completed
-

DOCUMENT CONTROL

Field	Value
Document Title	MarketFlow Enterprise ERP — System & Prototype Documentation
Application Version	2.4.0
Document Type	System & Prototype Documentation
Primary Domain	Digital Marketing Operations
Architecture	Modular Client-Side ERP
Database	PostgreSQL / Supabase
Access Control	Five-Pillar RBAC
Status	Prototype / Demonstration
Repository	MARKETFLOW-ENTERPRISE
Live Prototype	marketflow-enterprise.lovable.app

MARKETFLOW ENTERPRISE ERP

End of Document